

Math 220B Final

Zih-Yu Hsieh

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1 ND

Problem 1. Let R be a commutative ring such that $R_{\mathfrak{m}}$ is a Noetherian ring for every maximal ideal $\mathfrak{m}$. Prove or disprove that R is Noetherian.

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Problem 2. Let A and B be Noetherian rings together with surjective ring homomorphisms $f : A \rightarrow C$ and $g : B \rightarrow C$. Let $D = A \times_C B = \{(a, b) \in A \times B \mid f(a) = g(b)\}$ be the subring of the direct product ring $A \times B$. Prove or disprove that D is a Noetherian ring.

Solution. Idea: It must be Noetherian.

Try to prove the map $A \times_C B \twoheadrightarrow A \twoheadrightarrow C$ has kernel being exactly $\ker(f) \times \ker(g)$, which it forms the following exact sequence as $A \times_C B$ -module:

$$0 \rightarrow \ker(f) \times \ker(g) \rightarrow A \times_C B \twoheadrightarrow C \rightarrow 0 \quad (2.1)$$

Notice that since $\ker(f) \subseteq A$ and $\ker(g) \subseteq B$ are finitely generated by Noetherian Property (say $\ker(f) = (a_1, \dots, a_n)$, $\ker(g) = (b_1, \dots, b_m)$), then, $(a_1, 0), \dots, (a_n, 0), (0, b_1), \dots, (0, b_m)$ generates $\ker(f) \times \ker(g)$: Here, it uses the fact that any $a \in A$ has some $b \in B$, such that $(a, b) \in A \times_C B$ (by surjectivity of f, g). As a result, any $k \in \ker(f), l \in \ker(g)$, there exists $r_1, \dots, r_n \in A, s_1, \dots, s_m \in B$, such that $k = r_1 a_1 + \dots + r_n a_n$ and $l = s_1 b_1 + \dots + s_m b_m$. Suppose $(r_1, p_1), \dots, (r_n, p_n), (q_1, s_1), \dots, (q_m, s_m) \in A \times_C B$, then we have the following:

$$\begin{aligned} & (r_1, p_1)(a_1, 0) + \dots + (r_n, p_n)(a_n, 0) + (q_1, s_1)(0, b_1) + \dots + (q_m, s_m)(0, b_m) \\ &= (r_1 a_1, 0) + \dots + (r_n a_n, 0) + (0, s_1 b_1) + \dots + (0, s_m b_m) \\ &= (r_1 a_1 + \dots + r_n a_n, s_1 b_1 + \dots + s_m b_m) \\ &= (k, l) \end{aligned} \quad (2.2)$$

Hence, $\ker(f) \times \ker(g)$ is a fin. gen. $A \times_C B$ -module. In particular, this proof is doable for any submodule $I \times J \subseteq \ker(f) \times \ker(g)$

Problem: the kernel is finitely generated doesn't imply every of its submodule is finitely generated (Here C has no problem, as it's already a Noetherian A -mod / B -mod, so it's naturally a Noetherian $A \times_C B$ -Mod)

3 ND (Type the idea up)

Problem 3. Let R be a commutative local ring with maximal ideal $\mathfrak{m}$. Assume that $\mathfrak{m}$ is principal and $\bigcap_{n \geq 1} \mathfrak{m}^n = (0)$. Prove or disprove that R is Noetherian.

Solution. Idea: R is not necessarily Noetherian. Consider $R = k[x_1, \dots]/(x_1^2, \dots)$ (which has infinite indeterminates), which R is clearly not Noetherian (by choosing $(\overline{x_1}) \subset (\overline{x_1}, \overline{x_2}) \subset \dots$ as the ascending chain).

Then, we claim that $\mathfrak{m} = (\overline{x_1}, \dots) \subseteq R$ is maximal.

On the other hand, each $\mathfrak{m}^n$ must be generated by $\overline{x_{i_1}} \dots \overline{x_{i_m}}$, where each i_j are distinct, and $m \geq n$ (since all the powers ≥ 2 for the same indeterminate gets killed by the quotient). So, the intersection $\bigcap_{n \geq 1} \mathfrak{m}^n = (0)$.

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Problem 4. Is there an integral domain R whose all maximal ideals are principal but R is not a PID? Justify your answer.

5 ND (Type the ideal up)

Problem 5. Let R be an integral domain in which the set of principal ideals satisfies the ascending chain condition. Prove or disprove that R is Noetherian.

Solution. Idea: In a UFD, its principal ideals satisfies the ascending chain condition (because of unique factorization, and each strict inclusion of principal ideals must decrease in the number of irreducible factors). So, to provide a counterexample, it suffices to provide a UFD that's not Noetherian.

EX: $k[x_1, \dots]$, since every polynomial can be included into a polynomial ring with finite indeterminate, then it necessarily has a factorization (and it is unique, simply because any two factorizations can be included into the same finite indeterminate polynomial rings, by choosing enough polynomials). So, it's a UFD, but on the other hand it's not Noetherian (consider $(x_1) \subset (x_1, x_2) \subset \dots$ a strict inclusion of ideal, or copy the proof that the ideal $(x_1, \dots)$ is not finitely generated).